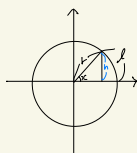




<인공지능 2주차 1차시 수업>

Math
- Tech
- 개념
- 기록록

$e = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n \rightarrow \frac{1}{n} \rightarrow 0, \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^{\frac{1}{n}} = 1$



$$l = r\theta$$

$$S = \frac{1}{2} l r = \frac{1}{2} r^2 \theta$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{h}{1-x} = \lim_{x \rightarrow 0} \frac{h}{x} = 1$$

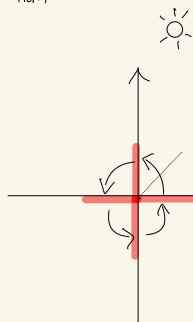
$$\lim_{x \rightarrow 0} \frac{x}{\sin x} = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1, \lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$$

$$\sin(\theta + \frac{\pi}{2}) = \cos \theta$$

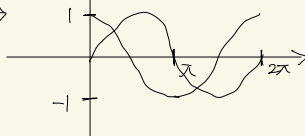
$$\cos(\theta - \frac{\pi}{2}) = \sin \theta$$

$$\sin(\theta + \pi) = -\sin \theta$$



파형은 도라치만

Real Term만 사용



① $\begin{array}{c} | \\ \hline 0 \end{array} \longrightarrow 1D$ scalar

② $\begin{array}{c} \longleftarrow | \longrightarrow \\ (-) \quad 0 \quad (+) \end{array} 1D$ Vector

③ $i^2 = -1, i^3 = i^2 \cdot i = -i, i^4 = 1$
↳ 모른다

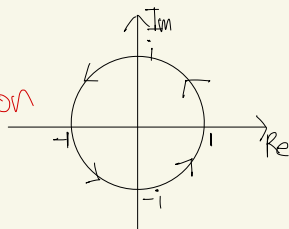
$\int \in \text{Real Number}$

실수: Real Number

허수: Imaginary Number

$$Z = a + bi: \text{복소수 (Complex)}$$

2D Rotation



부정적분

$$\int f(x) dx = F(x) + C$$

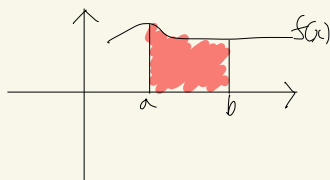
$$\text{ex) } \int \frac{1}{x} dx = \ln x + C, \int \cos x dx = \sin x$$

정적분

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\int \sin x dx = -\cos x, \int \sec^2 x dx = \tan x$$

$$\int \sec(x^2) \tan(x^2) 2x dx = \sec x^2 + C$$



$$y = \ln(\tan(\cos x))$$

$$\int \frac{-\sec^2(\cos x) \cdot \sin x}{\tan(\cos x)} dx = \ln|\tan(\cos x)| + C$$

$$y = \ln(\sec(e^{hx})) \quad y' = \frac{\sec(e^{hx}) \cdot \tan(e^{hx}) \cdot e^{hx} \cdot \frac{1}{x}}{\sec(e^{hx})} \downarrow dx$$

$$\int \frac{5x^4 + 6x + \frac{\cos x}{\sin x}}{x^5 + 3x^2 + \ln(\sin x)} dx = \ln|x^5 + 3x^2 + \ln(\sin x)| + C$$

$$\int f \cdot g dx = f \cdot G - \int f' \cdot G dx \quad / \quad \int x \sin dx = -\cos x + \int \cos x dx$$

f · 미분 편한거

g · 적분 "

$$f = x, f' = 1$$

$$g = \sin x \quad G = -\cos x$$

$$= -x \cos x + \sin x + C$$

$$= -\cos x + (-x)(-\sin x)$$

$$+ \cos x + 0$$

$$= x \sin x$$

100, 200, 300

$$\bar{x} = M_x = E[X_i]$$

$$Avg = \frac{\sum_{i=1}^N x_i}{N}$$

100	평차
200	-100 → 0 ²
300	0
	100

평균 편차: 0

음 → 양

① 1

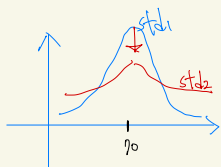
② 0

③ e

$$\sqrt{Var} = \text{표준편차 (Standard Deviation)} = \text{std}$$

Variance (분산): 편차의 제곱을 평균

$$Var = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N} \quad \text{평균}$$



std1 < std2 → 얼마나만큼 퍼졌나.

$$y = f(t)$$

$$t=0, 100 \text{ 마리}$$

$$t=1, 300 \text{ 마리}$$

$$t=0, ?$$

$$\frac{df(t)}{dt} = k \cdot f(t)$$

↓
1차 미분방정식

$$\frac{df(t)}{dt} = k \cdot f(t) \Rightarrow \frac{df(t)}{f(t)} = k \cdot dt$$

$$\ln|f(t)| = kt + C$$

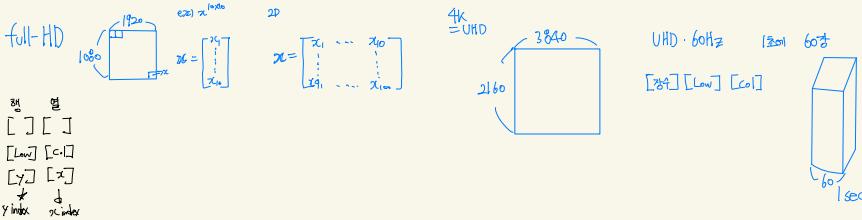
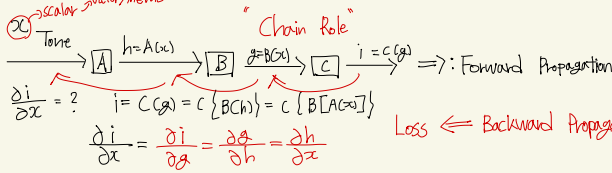
$$|f(t)| = e^{kt+C} = e^{kt} \cdot e^C$$

$$f(t) = \pm e^C \cdot e^{kt} = ce^{kt}$$

$$f(t) = ce^{kt}$$

$$\int \frac{1}{f} df = \int k dt \quad \leftarrow \ln|f| + C_1 = kt + C_2$$

<인공지능 2주차 2차시 수업>



X 인덱스

$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

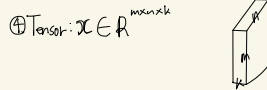
$|x| = \sqrt{2}$
 $\angle x = \tan^{-1}(1) = \frac{\pi}{4}$

① scalar : $x \in \mathbb{R}$

② Vector : $x \in \mathbb{R}^n \Rightarrow x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \Rightarrow x_i \in \mathbb{R} \quad (i=1 \sim n)$

③ Matrix : $x \in \mathbb{R}^{m \times n} \Rightarrow x = \begin{bmatrix} x_{11} & \dots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \dots & x_{mn} \end{bmatrix} \Rightarrow x_{ij} = \begin{bmatrix} i=1 \sim m \\ j=1 \sim n \end{bmatrix}$

$= [x_1 \dots x_n], x_n \in \mathbb{R}^m$
 $k=1 \sim n$



① Unit Vector $x \in \mathbb{R}^2 \quad x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \|x\|_2 = 5 \Rightarrow \|x\|_2 = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$ Norm

② Base Vector $x \in \mathbb{R}^n \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

$w_1 x_1 + w_2 x_2$
= Weighted Sum
= Linear Combination (선형 조합)

Vector space

closed

1 2 3 ... (개수)
1 1 1 ... (모든)

1 $\rightarrow$ 77 (7개) Linear Dependent
1 $\rightarrow$ 1 (1개) Linear Independent
고유한 기저 모임을 Basis Vector라고 부름

$x_1 \rightarrow \mathbb{R}^2 = \vee \quad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$w_1 x_1 + w_2 x_2 \Rightarrow \text{Span}$

$\text{Span}(x_1, x_2) = \{w_1 x_1 + w_2 x_2 \mid w_1, w_2 \in \mathbb{R}\}$
Linear C

$x_1 \cup x_n$

$w_1 x_1 + w_2 x_2 \dots + w_n x_n$

$w_1 \dots w_n = 0$ 선형 독립

$w_1 x_1 + w_2 x_2 = 0$

$w_1 x_1 = -w_2 x_2$
 $\downarrow$
 w_2

$\|x\|_2^2 = \sum x_i^2 = \|x\|^2$

$x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$
 $\downarrow$
Rank 1 $\Rightarrow$ (Basis Vector의 개수)

$x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
 $\downarrow$
Rank 3 $\Rightarrow$ full Rank

full Rank $\Rightarrow$ unique inverse matrix
ex) if (k != full Rank)
 $k^{-1} = \text{None}$

<인공지능 3주차 1차시 수업>

Col $x \in \mathbb{R}^m$

Metric $x \in \mathbb{R}^{m \times n} \Rightarrow x =$

Tenst $x \in \mathbb{R}^{m \times n \times k}$

Unit Vector $\rightarrow \pi_i \rightarrow$ Norm

$x \in \mathbb{R}^m$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$$

$$l_1 = \sum |x_i| = \|x\|_1$$

$$l_2 = \sqrt{\sum x_i^2} = \|x\|_2$$

$$ex) x = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$l_1 = 10$$

$$l_2 = \sqrt{3^2 + (-2)^2} = \sqrt{13}$$

Supervised Learning

A (정답) Label (Ground truth)

B (예측)

$$\sum_{i=1}^n |A_i - B_i| = L_1 \text{ Loss Function}$$

$$\text{Loss} = \sqrt{\sum_{i=1}^n (A_i - B_i)^2} = L_2$$

Parameter $\theta \rightarrow$ 최솟값은 만들어주는 것

Arg min Loss Function

$$= \arg \min_{\theta} \left(\sum_{i=1}^n (A_i - B_i) \right)$$

Basis

Span = Linear Combination (선형 조합)

Vector space $\cdot 0$ (Basis 1) zero

Vector space x

$$x_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} : x_1, x_2 \text{ Linear Independent}$$

$$\therefore \text{Span} \{x_1, x_2\} = \{w_1 x_1 + w_2 x_2 \mid w_1, w_2 \in \mathbb{R}\} \in \mathbb{R}^2 = \checkmark$$

$\Rightarrow$ Basis Vector $\approx$ Span $\approx$ zero Vector space $\approx$ Close $\approx$ Slope $\approx$ Orth.

$$\begin{bmatrix} 3 \\ 0 \end{bmatrix} = \frac{3}{2} \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} = w_1 \begin{bmatrix} 2 \\ 0 \end{bmatrix} + w_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{cases} 2w_1 + w_2 \\ w_2 \end{cases}$$

$$2w_1 + 4 = 3 \quad w_1 = -\frac{1}{2} \quad w_2 = 4$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

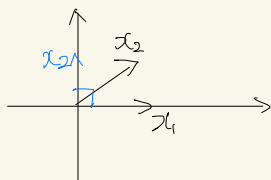
$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Span} (x_1, x_2) \in \mathbb{R}^2 = \checkmark$$

$$\text{Span} (x_1, x_2, x_3) \in \mathbb{R}^3$$



$\approx$ orthogonal

$$w_1 x_1 + w_2 x_2 = w_3 x_3$$

L. indep $\xrightarrow{(0) \text{ (x)}} \text{orthogonal}$
 $\xleftarrow{0}$

Shall <Basis Vector $\approx$ Linear Indep>
 $\approx$ have to $\approx$ must

$$\mathbb{R}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbb{R}^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$x_1 \quad x_2 \quad x_3$

$$x_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \text{orthonormal (직각의 크기 1)}$$

∴ Basis Vector는 span을 이 Vector space를 생성한다.
 → Basis vector는 Linear Independent이다.

$$A \in \mathbb{R}^{m \times n}$$

$$AB \Rightarrow C \in \mathbb{R}^{m \times k}$$

$$\begin{pmatrix} m \times n \\ \downarrow \\ n \times k \end{pmatrix}$$

① Transpose

$$A \in \mathbb{R}^{m \times n}, A^T \in \mathbb{R}^{n \times m}$$

$$\left\{ \begin{matrix} m \times n & n \times m & m=n \\ \left[\begin{matrix} \\ \end{matrix} \right] & \left[\right] & \left[\right] \\ \text{"rectangular"} & & \text{"square"} \end{matrix} \right\}$$

$$(AB)^T = B^T \cdot A^T$$

$$(A^T)^T = A$$

$$\sqrt{(A+B)^T = A^T + B^T}$$

② Symmetry (대칭성)

$$A = A^T$$

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$$

$$A = XX^T$$

$$A = A^T$$

$$(XX^T)^T = (X^T)^T \cdot X^T = XX^T$$

③ Inverse (역행렬)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

full Rank(x)=2 / Linear Independent

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \text{Rank}(A)=3$$

full Rank를 가지는 Inverse Matrix 존재.

⇒ 3 Basis vector에 Linear Independent이다.

$$AB = I \rightarrow \text{정렬행렬 (Identity)} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$B = A^{-1}$$

$$(AB)^T = B^T \cdot A^T$$

$$(A^{-1})^{-1} = A$$

$$\sqrt{(A+B)^{-1} \neq A^{-1} + B^{-1}}$$

대각행렬

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Zero Matrix

$$x \in \mathbb{R}^m = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}, y \in \mathbb{R}^n = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \quad n=10, m=5 \quad z \in \mathbb{R}^n$$

$$z_1 = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + \dots + w_nx_n$$

$$= y = W^T x \in \mathbb{R}^{n \times 1}$$

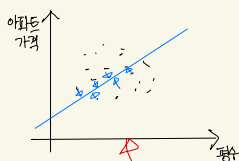
$$\downarrow$$

$$(n \times m) (m \times 1)$$

$$z = W^T \cdot x$$

$$= W^T \cdot W^T x$$

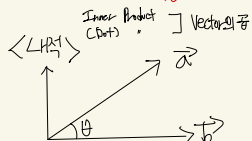
<인공지능 3주차 2차시 수업>



x_1, x_2
Linear regression (선형회귀)

$$Ax=b \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b \end{bmatrix}$$

$Ax=b$
 $A^T \cdot A = A^T \cdot b$
square
 $(n \times n)(n \times n) \quad \therefore x = (A^T \cdot A)^{-1} \cdot A^T \cdot b$



$$\vec{a} \cdot \vec{b} = \langle \vec{a}, \vec{b} \rangle = \|\vec{a}\|_2 \|\vec{b}\|_2 \cos \theta$$

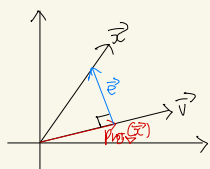
$\theta = \frac{\pi}{2} = \text{orthogonal}$

$$R^{n \times 1} \vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad x = R^n, y = R^n$$

$\langle x, y \rangle = x^T \cdot y$
 $(n \times 1)(n \times 1) \rightarrow (n \times 1)(n \times 1)$
 $= \sum_{i=1}^n x_i y_i$
 $(x_1, \dots, x_n) \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

$$\langle x, x \rangle = x^T \cdot x = \sum_{i=1}^n x_i^2 = \|x\|_2^2 \quad \therefore \|x\|_2^2 = x^T x$$

② projection



i) $p_{proj} \vec{x} = \alpha \cdot \vec{v} \quad (\alpha \in R)$
 ii) $p_{proj} \vec{x} + \vec{e} = \vec{x} \quad \therefore \vec{e} = \vec{x} - p_{proj} \vec{x}$
 iii) $\vec{e} \perp \vec{v} \quad \therefore \langle v, e \rangle = \langle e, v \rangle = 0$

$$\begin{aligned} &= v^T e \\ &= v^T (x - p_{proj}) \\ &= v^T x - v^T p_{proj} \\ &= v^T x - \alpha v^T v = 0 \quad \therefore \alpha = \frac{v^T x}{v^T v} \end{aligned}$$

$$p_{proj} \vec{x} = \alpha \vec{v} = \frac{v^T x}{v^T v} \cdot v = \frac{v v^T}{v^T v} x = \frac{v v^T}{\|v\|_2^2} x$$

Inner Product

③ 벡터 편미

① $y = Ax, y \in R^m, A \in R^{m \times n}, x \in R^n$

$$\Rightarrow \frac{\partial y}{\partial x} = \frac{\partial Ax}{\partial x} = A$$

$y = Ax$

$$\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$\begin{aligned} y_1 &= a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ y_2 &= a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ &\vdots \\ y_m &= a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \end{aligned}$$

② $\alpha = y^T A x, y \in R^m, A \in R^{m \times n}, x \in R^n$

i) $\frac{\partial \alpha}{\partial x} = y^T \cdot A$
 ii) $\frac{\partial \alpha}{\partial y} = \frac{\partial y^T A x}{\partial y} = \frac{\partial (y^T A)}{\partial y} x = \frac{\partial (A^T y)}{\partial y} = x^T \cdot A^T$

$$\hookrightarrow \frac{\partial y^T A x}{\partial y^T} = (A x)^T = x^T \cdot A^T$$

$$y_i = a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n$$

$(i=1, \dots, m) \quad \frac{\partial y_i}{\partial x_j} = a_{ij}$

$$\Rightarrow y_1 = f(x_1, x_2, \dots, x_n) \Rightarrow \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \dots & \frac{\partial y_1}{\partial x_n} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \dots & \frac{\partial y_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \frac{\partial y_m}{\partial x_2} & \dots & \frac{\partial y_m}{\partial x_n} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = A$$

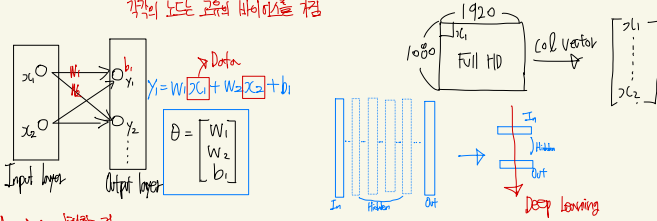
Jacobian

③ $\alpha = x^T A x, A \in R^{n \times n} \text{ (square)}$

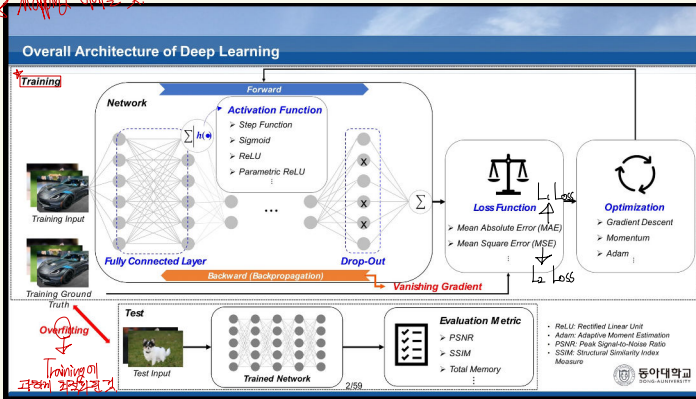
$$\begin{aligned} \frac{\partial \alpha}{\partial x} &= \frac{\partial (x^T A x)}{\partial x} = \frac{\partial (x^T A x)}{\partial x^T} + \frac{\partial (x^T A x)}{\partial x} \\ &= (A x)^T + x^T \cdot A \\ &= x^T \cdot A^T + x^T \cdot A = x^T (A + A^T) \end{aligned}$$

<인공지능 4주차 이론>

Perceptron (인간의 뉴런을 모사) 다중입력 → 단일출력
각각의 노드 코어의 바이어스를 곱함



★ Applying 이론화 것



Learning

Supervised learning (지도학습)

⇒ 샘플이 있다 ⇒ 답이 있다

↳ Label, Ground Truth

Training Data ↔ Test Data

ex) 95%

85%

90%

90% ✓ Test 단계에서 정확도가 높으면 좋음

<논리회로 학습법>

기호적인 인공신경망
(컴퓨터를 이해하고 판단할
것의 원리 이해)

ex) Pixel
고양이의 RGB 값과 주변 배경의 RGB 값이
같으면 고양이로 구분 못하는 오류가 생긴

↓
XOR 문제

인공적인 인공신경망
(인공 인공신경망
학습 방법)

정: Inference

Supervised learning

→ Regression (Output: Real Number)

→ Classification

Binary Classification (Output: 0 or 1)

Multi-Label (Output: 0, 1, 2, ..., N)

$x_1 \circ w_1 \rightarrow x_2 \circ w_2 \rightarrow x_3 = w_1x_1 + w_2x_2 + b_1$ (ex) 밝이 강한 곳에서 빛을 조금 줄여가게끔 하겠다

$x_2 \circ w_2$ 차이를 구별할 수 있나? → NO

Sigmoid = $\frac{1}{1 + e^{-x}}$



step function

이것이 해결과 같은 것은 이런 Network가 있을 때
Perceptron을 통해서 w값과 bias를 구해준다.
어떻게? Loss를 최소화할 수 있는 방법으로
어떤 방법? Backpropagation으로
Overfitting을 유도하면서.

Logic gate



0	0	0
0	1	0
1	0	0
1	1	1



0	0	0
0	1	1
1	0	1
1	1	1



0	1
1	0



0	0	1
0	1	1
1	0	1
1	1	0



0	0	1
0	1	0
1	0	0
1	1	0



0	0	0
0	1	1
1	0	1
1	1	0

wt. x
~~($m \times n$)~~ ($m \times n$)

$$\text{loss } \sum_{i=1}^n g^2$$

($\mathbb{R} \times \mathbb{R}$)
 $\mathbb{R}$

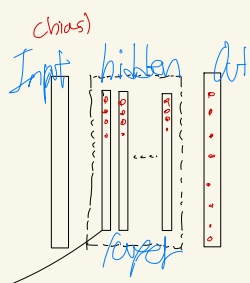
$$\begin{aligned} x \in \mathbb{R}^m \quad g \in \mathbb{R}^n \quad b \in \mathbb{R}^n \\ \hline w \in \mathbb{R}^{m \times n} \\ w^T \in \mathbb{R}^{n \times m} \end{aligned}$$

$$\frac{\partial \mathcal{L}}{\partial g} = 2g \in \mathbb{R}^n$$

<인공지능 시작 이론>

MLP (Multi Layer Perception)

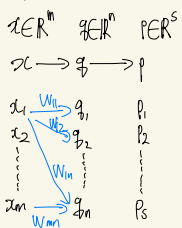
$y = W^T x$
 $x \in \mathbb{R}^n$ $W \in \mathbb{R}^n$
 $= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$ $= \begin{bmatrix} W_{11} \\ \vdots \\ W_{n1} \end{bmatrix}$
 $W^T x \quad C(xn)(nx1) = C(1x1)$
 $y = W^T x + b \rightarrow \Delta$ 증가
 $y = h(W^T x + b)$



MLP
 = fully connected layer
 = FC layer

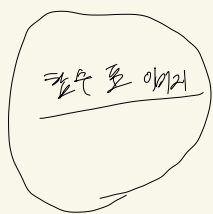
N.P : $W \cdot b$
 H.P : 사람이 결정
 Opt.
 Best-effort (노력)

ex) Hidden layer 구조를 살펴볼게.



$i) g = h[W^T x + b_1] \in \mathbb{R}^n$
 $ii) p = h[W_2^T g + b_2] \in \mathbb{R}^s$
 $iii) p = h[W_3^T [h[W^T x + b_1] + b_2]$

활성화 함수 (Activation function)

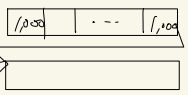
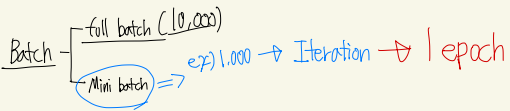
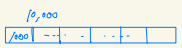


만일 이 년 것과 원본이 같아 적은 학습을 줌
 $Loss = \sum |y - \hat{y}| = L_1 Loss$
 $= \sum (y_i - \hat{y}_i)^2 = L_2 Loss$
 무작위 값이 경계값이 없을 때 Best-effort
 함께 따라 L1, L2 사용이 일반
 Training이 편하다.
 Loss가 낮아지면 학습
 W, b 같은 Parameters 값이 적절함
 Loss가 적으면 학습이 잘 됨, 결과적으로

$L_1 Loss = MAE$ (평균 절대 오차)
 $L_2 Loss = MSE$ (평균 제곱 오차)
CEE (Cross Entropy)
 Loss를 줄이기 위해
 y, y' CEE
 원 y와 예측한 y'의 Entropy
 → y와 y'의 차이 (Distance)를 줄여야 함
 → y와 y'의 차이 (Distance)를 줄여야 함

Network parameter = ?

- Forward Propagation
- Backward Propagation



Full batch (10,000)을 Mini batch (1,000)로 나눠서 Iteration 하면 10번이고
 10번의 Iteration이 끝나면 1 epoch이라고 함.
 0/1 Mini batch, epoch은 Hyper parameter 즉, 사람이 정함